

Probability – II: Assignment 2

Indian Statistical Institute, Delhi

Utkarsh Bharadwaj | BSD-BG-2521

October 7, 2026

Exercise 1.

Problem [1 mark]

X is a nonnegative random variable with finite second moment. Show that

$$\mathbb{P}(X > 0) \geq \frac{\mathbb{E}^2(X)}{\mathbb{E}(X^2)}.$$

Solution

Consider the alternative form of the Cauchy–Schwarz inequality

$$\mathbb{E}^2(XY) \leq \mathbb{E}(X^2) \mathbb{E}(Y^2).$$

Let

$$Y = \mathbb{1}_{\{X > 0\}} = \begin{cases} 1 & \text{if } X > 0 \\ 0 & \text{otherwise.} \end{cases}$$

This implies $\mathbb{E}(Y^2) = \mathbb{P}(X > 0)$ and $\mathbb{E}^2(XY) = \mathbb{E}^2(X)$ for $X > 0$. Hence

$$\mathbb{P}(X > 0) \geq \frac{\mathbb{E}^2(X)}{\mathbb{E}(X^2)}. \quad \blacksquare$$

Exercise 2.

Problem [1 mark]

We can define a set called $\{A_n \text{ e.v.}\}$ (where e.v. is the abbreviation of eventually) as

$$\{A_n \text{ e.v.}\} = \bigcup_n \bigcap_{k \geq n} A_k.$$

Let $\mathbb{P}\{A_n \text{ e.v.}\} = 1$. Show that $\mathbb{P}(A_n) \rightarrow 1$.

Solution

Given $\{A_n \text{ e.v.}\} = \bigcup_n \bigcap_{k \geq n} A_k$. Define $B_n = \bigcap_{k \geq n} A_k$. Thus

$$\{A_n \text{ e.v.}\} = \bigcup_n B_n.$$

It is also quite evident that

$$(a) \ B_1 \subseteq B_2 \subseteq \dots$$

$$(b) \ B_n \subseteq A_n \quad \forall n$$

$$\implies \mathbb{P}(A_n \text{ e.v.}) = \mathbb{P}\left(\bigcup_{n \geq 1} B_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(B_n) = 1 \quad (\text{Given})$$

From (b), we have $\mathbb{P}(B_n) \leq \mathbb{P}(A_n)$

$$\implies \lim \mathbb{P}(B_n) \leq \lim \mathbb{P}(A_n) \implies \lim \mathbb{P}(A_n) \geq 1 \implies \lim \mathbb{P}(A_n) = 1$$

$$\implies \mathbb{P}(A_n) \rightarrow 1. \quad \blacksquare$$

Exercise 3.

Problem [1 mark]

Consider $[-1, 1]$ as a 1-dimensional dart board, 0 being the bull's eye. A drunkard is throwing darts at this board hitting the board at any random points (all points equally likely). After each attempt he gets a prize if he hits within distance $\frac{1}{n}$ of the bull's eye (if it is the n -th attempt). What is the chance that he gets infinitely many prizes? [Assume infinite life for the guy, but don't drink to it!]

Solution

Let A_n be the event that he gets a prize in the n^{th} attempt.

$$\mathbb{P}(A_n) = \frac{\text{Length of } \left[-\frac{1}{n}, \frac{1}{n}\right]}{\text{Length of } [-1, 1]} = \frac{2/n}{2} = \frac{1}{n}.$$

Clearly, $\sum \mathbb{P}(A_n) = \sum \frac{1}{n} \rightarrow \infty$.

$\Rightarrow$ From the 2nd Borel–Cantelli Lemma, we have $\mathbb{P}(A_n \text{ i.o.}) = 1$.

Exercise 4.

Problem [1 mark]

If $X_n \xrightarrow{P} X$, show that $|X_n| \xrightarrow{P} |X|$. Prove it directly using the definition (and not the continuous mapping theorem).

Solution

Given $X_n \xrightarrow{P} X$. This implies $\forall \varepsilon > 0, \mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$.

Consider the reverse triangle inequality

$$||X_n| - |X|| \leq |X_n - X|.$$

Thus,

$$\left\{ ||X_n| - |X|| > \varepsilon \right\} \subseteq \left\{ |X_n - X| > \varepsilon \right\} \quad \forall \varepsilon > 0$$

$$\implies \mathbb{P}\left(||X_n| - |X|| > \varepsilon \right) \leq \mathbb{P}\left(|X_n - X| > \varepsilon \right) \rightarrow 0$$

$$\implies \lim \mathbb{P}\left(||X_n| - |X|| > \varepsilon \right) \leq 0 \implies \mathbb{P}\left(||X_n| - |X|| > \varepsilon \right) \rightarrow 0 \quad \forall \varepsilon > 0$$

$$\implies |X_n| \xrightarrow{P} |X|. \quad \blacksquare$$

Exercise 5.

Problem [1 mark]

If $X_n \xrightarrow{d} 5$, then show that $X_n \xrightarrow{P} 5$.

Solution

Let $X = 5$ be a degenerate r.v. Thus, its CDF is

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 1 & \text{if } x \geq 5 \\ 0 & \text{if } x < 5. \end{cases}$$

$F_X(x)$ is continuous $\forall x \in \mathbb{R} \setminus \{5\}$.

Given $X_n \xrightarrow{d} 5$. Thus, by definition of convergence in distribution,

$$\lim_{n \rightarrow \infty} F_n(x) = F_X(x) \quad \forall x \neq 5$$

where $F_n(x) = \mathbb{P}(X_n \leq x)$ is the CDF of X_n .

TST: $\forall \varepsilon > 0, \lim \mathbb{P}(|X_n - 5| > \varepsilon) = 0$.

$$\begin{aligned} \implies \mathbb{P}(|X_n - 5| > \varepsilon) &= \mathbb{P}(X_n > 5 + \varepsilon) + \mathbb{P}(X_n < 5 - \varepsilon) \\ &= 1 - F_n(5 + \varepsilon) + F_n(5 - \varepsilon) \end{aligned}$$

$$\begin{aligned}
\Rightarrow \quad \lim \mathbb{P}(|X_n - 5| > \varepsilon) &= 1 - \lim F_n(5 + \varepsilon) + \lim F_n(5 - \varepsilon) \\
&= 1 - F_X(5 + \varepsilon) + F_X(5 - \varepsilon) \\
&= 1 - 1 + 0 = 0 \\
\Rightarrow \quad \lim \mathbb{P}(|X_n - 5| > \varepsilon) &= 0 \quad \forall \varepsilon > 0 \\
\Rightarrow \quad X_n &\xrightarrow{P} 5. \quad \blacksquare
\end{aligned}$$

Exercise 6.

Problem [1 mark]

(DCT) Let (X_n) , X be random variables such that $X_n \xrightarrow{a.s.} X$ and $|X_n| \leq Y \forall n \in \mathbb{N}$ for some random variable Y with $\mathbb{E}(Y) < \infty$. Then show that $X_n \xrightarrow{L^1} X$.

Solution

Given $X_n \xrightarrow{a.s.} X$ and $|X_n| \leq Y$ a.s. $\forall n$. This implies

$$|X| = \lim_{n \rightarrow \infty} |X_n| \leq Y \quad \text{a.s.}$$

By triangle inequality, we have

$$|X_n - X| \leq |X_n| + |X| \leq Y + Y = 2Y \quad \text{a.s.}$$

Consider $Z_n = 2Y - |X_n - X| \geq 0 \forall n \in \mathbb{N}$. Thus, we can apply Fatou's Lemma on Z_n

$$\mathbb{E}(\liminf Z_n) \leq \liminf \mathbb{E}(Z_n) \quad \dots \textcircled{1}$$

$$\Rightarrow \quad \liminf Z_n = \lim (2Y - |X_n - X|) = 2Y \quad \text{a.s.}$$

$$\mathbb{E}(\liminf Z_n) = 2\mathbb{E}(Y)$$

$$\begin{aligned}
\Rightarrow \quad \liminf \mathbb{E}(Z_n) &= \liminf (2\mathbb{E}(Y) - \mathbb{E}|X_n - X|) \\
&= 2\mathbb{E}(Y) - \limsup \mathbb{E}|X_n - X|
\end{aligned}$$

$\implies$ From ①, we have

$$2\mathbb{E}(Y) \leq 2\mathbb{E}(Y) - \limsup \mathbb{E}|X_n - X|$$

$$\implies \limsup \mathbb{E}|X_n - X| \leq 0 \implies \liminf \mathbb{E}|X_n - X| \geq 0$$

Thus:

$$\lim_{n \rightarrow \infty} \mathbb{E}|X_n - X| = 0 \implies X_n \xrightarrow{L^1} X. \quad \blacksquare$$

Exercise 7.

Problem [1+1 marks]

Let (X_n) be iid $\text{Poi}(\lambda)$. Let $Y_n = \frac{1}{n} \sum_{k=1}^n X_k^3$. Show that (Y_n) converges almost surely. Find the limit.

Solution

Let $W_k = X_k^3 \forall 1 \leq k \leq n$. The W_k are iid because X_k were iid. By Strong Law of Large Numbers, if $\mathbb{E}(|W_1|) < \infty$ then

$$Y_n = \frac{1}{n} \sum X_k^3 = \frac{1}{n} \sum W_k \xrightarrow{a.s.} \mathbb{E}(W_1) = \mathbb{E}(X_1^3).$$

Thus, $Y_n \xrightarrow{a.s.} \mathbb{E}(X_1^3)$.

Now, consider

$$\mathbb{E}(X_1) = \lambda, \quad \mathbb{E}(X_1(X_1 - 1)) = \lambda^2 \implies \mathbb{E}(X_1^2) = \lambda^2 + \lambda$$

$$\mathbb{E}(X_1(X_1 - 1)(X_1 - 2)) = \lambda^3 \implies \mathbb{E}(X_1^3) = \lambda^3 + 3\mathbb{E}(X_1^2) - 2\mathbb{E}(\lambda)$$

$$= \lambda^3 + 3\lambda^2 + \lambda \implies Y_n \xrightarrow{a.s.} \lambda^3 + 3\lambda^2 + \lambda.$$

Exercise 8.

Problem [2 marks]

Let $X_1, X_2, \dots$ be iid with some distribution having finite moments $\mathbb{E}(X_1^k) = \mu_k$ for $k = 1, 2, 3, 4$. We are interested in the sample CV of $X_1, \dots, X_n$:

$$CV_n = \frac{S_n}{\bar{X}_n},$$

where $\bar{X}_n$ and S_n^2 are the sample mean and variance of $X_1, \dots, X_n$. Use the multivariate CLT and delta method to obtain the asymptotic distribution of CV_n .

Solution

Let $T_n = \begin{pmatrix} \bar{X}_n \\ \frac{1}{n} \sum X_i^2 \end{pmatrix}$. Thus, by multivariate CLT

$$\sqrt{n} \left(T_n - \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \right) \xrightarrow{d} N_2(0, \Sigma) \quad \text{where} \quad \Sigma = \begin{pmatrix} \mu_2 - \mu_1^2 & \mu_3 - \mu_1\mu_2 \\ \mu_3 - \mu_1\mu_2 & \mu_4 - \mu_2^2 \end{pmatrix}.$$

→ Define $g(t_1, t_2) = \frac{\sqrt{t_2 - t_1^2}}{t_1}$, so $g(T_n) \approx CV_n$ and

$$g(\mu_1, \mu_2) = \frac{\sigma}{\mu_1} = cv$$

$$\Rightarrow \nabla g = \begin{pmatrix} -\frac{\mu_2}{\mu_1^2 \sigma} \\ 1 \\ 2\mu_1 \sigma \end{pmatrix} = \frac{1}{2\mu_1^2 \sigma} \begin{pmatrix} -2\mu_2 \\ \mu_1 \end{pmatrix} \quad \text{where } \sigma = \sqrt{\mu_2 - \mu_1^2}.$$

Thus, by Delta method, we have

$$\sqrt{n} (CV_n - cv) \xrightarrow{d} N(0, \sigma_{cv}^2) \quad \text{where} \quad \sigma_{cv}^2 = (\nabla g)^T \Sigma (\nabla g)$$

$$\Rightarrow \sigma_{cv}^2 = \frac{1}{4\mu_1^4 \sigma^2} \begin{pmatrix} -2\mu_2 & \mu_1 \end{pmatrix} \begin{pmatrix} \mu_2 - \mu_1^2 & \mu_3 - \mu_1\mu_2 \\ \mu_3 - \mu_1\mu_2 & \mu_4 - \mu_2^2 \end{pmatrix} \begin{pmatrix} -2\mu_2 \\ \mu_1 \end{pmatrix}$$

$$\sigma_{cv}^2 = \frac{4\mu_2^3 - \mu_1^2\mu_2^2 - 4\mu_1\mu_2\mu_3 + \mu_1^2\mu_4}{4\mu_1^4(\mu_2 - \mu_1^2)}$$

$$\Rightarrow \sqrt{n} \left(CV_n - \frac{\sqrt{\mu_2 - \mu_1^2}}{\mu_1} \right) \xrightarrow{d} N \left(0, \frac{4\mu_2^3 - \mu_1^2\mu_2^2 - 4\mu_1\mu_2\mu_3 + \mu_1^2\mu_4}{4\mu_1^4(\mu_2 - \mu_1^2)} \right). \quad \blacksquare$$